

Sync Creator Kit Updater

A batch synchronization engine that propagates rule and parameter changes across every Creator Kit in the Coding5s framework — in seconds, without disturbing a single style, formula, or format.

PILLAR: 1 — CODING5S CREATOR KITS RUNTIME: PYTHON 3.8+ · WINDOWS COM SCOPE: PYTHON · ELIXIR · GLEAM · DART

EXECUTIVE SUMMARY

The **Sync Creator Kit Updater** is the operational backbone that keeps every Creator Kit across the Coding5s ecosystem aligned to a single source of truth. As instructional rules, system prompts, and atomic framework parameters evolve, this engine eliminates the need to manually open and edit dozens of files across multiple programming-language tracks — collapsing an hours-long, error-prone task into a single one-click, format-preserving batch operation.

By writing static values directly through Windows COM Automation and forcing a full recalculation immediately after ingestion, the updater guarantees that every dependent formula inside each Creator Kit reflects the new ruleset instantly — with zero impact on fonts, borders, or conditional formatting.

5
PIPELINE STAGES

4+
LANGUAGE TRACKS

1
SOURCE OF TRUTH

0
FORMAT LOSS

EXECUTION PIPELINE

STAGE 01 Master Source Validation		✓ PRE-FLIGHT CHECK
OBJECTIVE Confirm the integrity of the update chain before touching a single Creator Kit file, guaranteeing the batch never runs against a missing or corrupted data source.	MECHANISM Verifies that <code>creator_kit_Base_File_for_CKs.xlsx</code> exists in the root folder; halts execution immediately with a critical error log if absent.	
KEY OUTPUTS • Fail-fast safety: no partial or silent batch runs • Clear diagnostics: critical error logged on failure		
STAGE 02 Recursive Discovery		os.walk()
OBJECTIVE Locate every target workbook across the full language-track hierarchy without requiring a manually maintained file list.	MECHANISM Recursively scans the root and all subdirectories (<code>python</code> , <code>elixir</code> , <code>gleam</code> , <code>dart</code>) for <code>.xlsx</code> files containing “Creator Kit”, excluding temp locks (<code>~\$</code>) and the base file itself.	
KEY OUTPUTS • Zero-maintenance discovery: new tracks are picked up automatically • Clean targeting: active-lock and base files are safely excluded		
STAGE 03 Master Matrix Extraction		COM READ
OBJECTIVE Capture the current ruleset from the single source of truth into memory, ready for propagation.	MECHANISM Launches background Excel via <code>win32com.client.Dispatch</code> , opens the base file, reads <code>RANGO_A_COPIAR</code> as a 2D array into RAM, then closes the base file immediately.	
KEY OUTPUTS • In-memory matrix: ready for high-speed batch writes • Minimal file lock time: base file closes right after read		

STAGE 04 Batch Ingestion & Recalculation

WRITE + CALC

OBJECTIVE

Propagate the updated matrix into every discovered Creator Kit while preserving 100% of existing visual formatting and formulas.

MECHANISM

For each file: opens in background Excel, maps target range dimensions to the source shape, writes `rango_destino.Value = matriz_valores`, forces `excel.Calculate()`, then saves and closes.

KEY OUTPUTS

• **Format-preserving injection:** styles, borders, conditional rules untouched

• **Instant formula sync:** every dependent cell recalculates in place

STAGE 05 Memory Cleanup & Release

FINALLY BLOCK

OBJECTIVE

Return the host machine to a clean state regardless of whether the batch succeeded or failed.

MECHANISM

Releases all COM objects, restores screen-updating and alert modes, and safely terminates Excel instances inside a guaranteed `finally` block.

KEY OUTPUTS

• **No orphaned processes:** Excel instances always terminate cleanly

• **Crash-safe:** cleanup runs even on mid-batch failure

TARGET MATRIX CONFIGURATION		
VARIABLE	VALUE	ROLE
ARCHIVO_BASE_NOMBRE	"creator_kit_Base_File_for_CKs.xlsx"	Central master source file name
PESTANA_BASE_ORIGEN	"BaseData"	Source sheet inside the base file
RANGO_A_COPIAR	"A1:B10"	Cell range extracted from the base file
PESTANA_DESTINO_CK	"FGen_S1"	Target sheet inside each Creator Kit
CELDA_INICIAL_DESTINO	"C95"	Top-left anchor cell for the destination paste

SYSTEM ARCHITECTURE

PILLAR_1/CODING5S_CREATOR_KITS/

creator_kit_Base_File_for_CKs.xlsx

Central data / rules source

creator_kit_updater.py

Core Python automation engine

Creator Kit Updater Runner.bat

1-click execution script

README.md

Documentation

python/

Language / topic subdirectory

└ Coding5s Python Core ... Creator Kit.xlsx

elixir/ gleam/ dart/

Additional language tracks

SETUP & EXECUTION WORKFLOW

SYSTEM REQUIREMENTS

• **OS:** Windows 10 / 11

• **Software:** Microsoft Excel installed locally

• **Environment:** Python 3.8+

• **Dependency:** `pip install pywin32`

OPERATIONAL SAFEGUARDS

• Native VBScript confirmation dialog before any overwrite

• Automatic exclusion of temp-lock (~\$) and base files

• Guaranteed COM cleanup on success or failure

• Console logging of every file processed in real time

CODING5S ECOSYSTEM · SYNC CREATOR KIT UPDATER · TECHNICAL FRAMEWORK BRIEF

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EXECUTION SEQUENCE

- 1

Configure update variables

Edit `PESTANA_BASE_ORIGEN`, `RANGO_A_COPIAR`, `PESTANA_DESTINO_CK` and `CELDA_INICIAL_DESTINO` in `creator_kit_updater.py`, then save.
- 2

Prepare the data source

Place the new rules or parameters inside `creator_kit_Base_File_for_CKs.xlsx`, within the specified range.
- 3

Execute the batch update

Double-click `Creator Kit Updater Runner.bat` and confirm the action in the prompt dialog.
- 4

Monitor console progress

Real-time log output reports each Creator Kit as it is located, loaded, updated, and recalculated.
- 5

Completion

Every Creator Kit now carries the updated rules with formulas fully refreshed — press any key to exit.

TECHNICAL FOOTER

KEY ARCHITECTURAL PILLARS

- **Single Source of Truth:** one master workbook drives every Creator Kit
- **COM-Native Automation:** direct Windows Excel automation via `win32com`
- **Recursive Discovery:** `os.walk()` across all language tracks
- **Format-Preserving Writes:** value-only injection, styles untouched

SYSTEM GUARANTEES

- **Zero style drift:** fonts, borders, conditional formatting stay intact
- **Full recalculation:** `excel.Calculate()` after every write
- **Fail-fast validation:** halts on missing master source
- **Guaranteed cleanup:** COM objects released in a `finally` block

REFERENCES

- `creator_kit_updater.py` — core engine
- `creator_kit_Base_File_for_CKs.xlsx` — master source
- `Creator Kit Updater Runner.bat` — execution script
- `README.md` — full documentation
- Dependency: `pywin32` (PyPI)